Comprehension Without Compliance: Harm Comprehension and Refusal Are Separately Encoded in Reasoning Models

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Abstract

Language models are trained to refuse harmful requests, but refusing a request is not the same as understanding that it is harmful. The distinction matters most for reasoning models, which think step by step before answering and which safety research increasingly relies on. When such a model refuses, is the refusal the result of reasoning about harm, or a reflex that runs alongside the reasoning without depending on it? We study where harm comprehension and refusal are represented inside three open reasoning models, including GPT-OSS-20B, whose ability to reason was trained by reinforcement learning rather than copied from a stronger model.

Harm comprehension and refusal turn out to be encoded separately, and they stay nearly independent of each other even in the model whose deliberation was genuinely trained. The internal representation of harm is real: it is strongly decodable, and in models that answer directly, steering it flips the model's stated judgment of whether a request is harmful. Yet in the reasoning models that representation is spread across the chain of thought and concentrated away from the point where the refusal is decided, and the refusal itself is not controlled by any single internal direction. The harm representation is related to, but distinct from, the moral-foundation structure that other work measures. We also report a practical obstacle: the standard test for whether such a direction is causal reads the model's stated judgment, and reasoning models do not state one cleanly, which limits how interpretability tools built for direct-answering models carry over.

Understanding that a request is harmful and acting on that understanding are separate things inside reasoning models, and they remain separate even when the model is trained to deliberate. This is a caution for safety methods that read a model's refusal as evidence of what it comprehends.

1 Introduction

Refusal in instruction-tuned language models can be removed by a single rank-one weight edit. Arditi et al. (Arditi et al., 2024) showed that refusal is mediated by one linear direction in the residual stream, and that orthogonalizing the model's weights against it suppresses refusal across many harmful requests. A prior paper in this series asked what that direction is made of, and found, in several dense base+instruct families, that the ablatable refusal direction is almost entirely residual against the six-foundation moral subspace: comprehension and compliance are separable, and refusal is the separable part (Reblitz-Richardson, 2026). Zhao et al. (Zhao et al., 2025) reached a parallel conclusion from a different angle, showing that harmfulness and refusal are encoded as distinct internal concepts at different token positions: harmfulness at the last instruction token (t_{inst}) and refusal at the last templated token ($t_{\text{post-inst}}$). Steering the harmfulness direction changes the model's

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belief that an instruction is harmful; steering the refusal direction toggles refusal without changing that belief.

Both results were established on models that answer directly. Reasoning models are different. They emit an extended chain of thought before answering, and at least part of the refusal decision is made inside that trace (Yamaguchi et al., 2025). A reasoning model that has been trained, by reinforcement learning, to deliberate about whether a request is harmful might route its refusal through that deliberation, re-coupling harm comprehension to the refusal decision that non-reasoning models keep separate. Whether genuine deliberative training re-couples comprehension and refusal, or whether the dissociation extends into the reasoning regime, is the question this paper studies.

We study it across three open reasoning models that isolate the deliberative axis. GPT-OSS-20B (OpenAI, 2025) learned to reason by reinforcement learning under deliberative alignment, the one model in which the chain of thought is a trained behavior rather than an imitated one. DeepSeek-R1-Distill-Llama-8B and DeepSeek-R1-Distill-Qwen-14B (DeepSeek-AI, 2025) acquired their reasoning by supervised distillation of R1 traces onto a non-reasoning base. We extract directions at t_{inst} and $t_{\text{post-inst}}$ with conventions held fixed across the panel (section 3), and ask where harm comprehension sits relative to the refusal decision in each.

We report four findings.

Finding 1: The harmfulness/refusal dissociation holds in reasoning models, deliberative training included. Harmfulness is strongly decodable at the instruction token (Cohen's $d' \approx 4.4$–5.0 across the panel) yet nearly orthogonal to the refusal direction (cosine 0.11–0.16), and this holds for GPT-OSS-20B, whose reasoning was learned by reinforcement learning, as cleanly as for the distilled models (Figure 1). Training a model to deliberate about harm does not fuse its harm comprehension with its refusal behavior.

Finding 2: Harm comprehension is distributed across the reasoning trace and displaced from the decision, and the refusal decision is itself distributed. GPT-OSS-20B carries the most trace-level moral content of the panel, and that content peaks near the first third of the chain of thought and falls to its lowest value at the decision (Figure 2). The refusal decision is not bottlenecked through any one direction: on a held-out set, no single direction, the refusal direction itself included, cleanly ablates GPT-OSS-20B refusal (Figure 3). The model that most clearly reasons about harm is the one whose refusal is least reducible to a single handle.

Finding 3: The harmfulness representation is causally real, but largely distinct from Moral Foundations. Steering the harmfulness direction flips a model's stated harm judgment on direct-answering instruct models, establishing that the direction is causal and not merely decodable (Figure 5). Yet it projects mostly outside the six-foundation moral subspace, overlapping it above chance but by only a small fraction (Figure 4). The harm-judgment that models route their refusal-comprehension through is related to, but not the same as, the moral foundations of Haidt (Graham et al., 2013).

Finding 4: Reasoning models do not expose a clean judgment readout, which constrains interpretability methods built on direct-answering models. The reply-inversion test that validates the harmfulness direction reads the model's stated harm judgment, and reasoning models do not state one cleanly: across three readout mechanisms they reason past the question, echo it, or over-judge when forced, while the same readout is clean on instruct models (section 3). The judgment is present in the model but not exposed in a position a readout can reach. As the growing body of refusal- and harmfulness-direction work (Arditi et al., 2024, Zhao et al., 2025) is applied to the reasoning models safety research increasingly depends on, the behavioral readout channel, not the direction, becomes the binding constraint.

We are explicit about what we do not find. The clean functional-versus-imitated split we sought, harm comprehension load-bearing for refusal in the RL-deliberative model and decorative in the distilled ones, is not cleanly runnable here. GPT-OSS-20B's refusal is distributed, so no low-dimensional ablation isolates a load-bearing test (Finding 2); and the distilled models barely refuse at baseline (Figure 6), a behavior confounded with R1 distillation degrading refusal training, so their failure to translate harm comprehension into refusal cannot license a claim that the comprehension is decorative. The honest contribution is the dissociation and its reach, extended into the reasoning regime and grounded against a peer-reviewed causal account of harmfulness encoding.

2 Related Work

Refusal as a linear direction. Arditi et al. (Arditi et al., 2024) showed that refusal in instruction-tuned models is mediated by a single residual-stream direction, removable by rank-one weight orthogonalization; the Heretic tool (p-e-w, 2025) automates the operation across families. That refusal is single-direction-ablatable is established. Our concern is where that direction sits relative to harm and moral representations, and whether, in reasoning models, the refusal decision is bottlenecked through one direction at all.

Harmfulness and refusal as separate concepts. Zhao et al. (Zhao et al., 2025) established, on instruct models, that harmfulness and refusal are encoded as distinct internal concepts at different token positions: harmfulness at the last instruction token, refusal at the last templated token. Steering the harmfulness direction changes the model's belief that an instruction is harmful, while steering refusal toggles the behavior without changing that belief, and adversarially finetuning a model to accept harmful instructions leaves the harmfulness representation largely intact. We build directly on this account, extend its token-position dissociation into the reasoning regime, position our Moral Foundations subspace against its harmfulness direction, and adopt its reply-inversion test as our causal yardstick. The point of contact with our distill result is its finetuning-robustness finding: a model whose refusal is degraded can retain its harm comprehension.

Refusal in reasoning traces. Yamaguchi et al. (Yamaguchi et al., 2025) showed, on DeepSeek-R1-Distill-Llama-8B among others, that reasoning models make part of the refusal decision inside the chain of thought rather than at the prompt-response boundary, that the opening sentence of the trace can determine the refusal outcome, and that linear refusal directions ablate harmful compliance less reliably on reasoning models than on non-reasoning ones. Our two-site extraction follows this localization; we add the harm-comprehension geometry and the distributed-decision result, the latter consistent with their weaker single-direction ablation.

Comprehension and compliance. A prior paper in this series found, across dense base+instruct families, that the ablatable refusal direction is almost entirely residual against the six-foundation moral subspace, and that ablating it removes refusal while leaving moral comprehension and behavioral judgment intact (Reblitz-Richardson, 2026). Comprehension and compliance are separable. This paper asks whether that separation survives in reasoning models whose refusal is partly deliberated, and finds that it does.

Moral Foundations in language models. Our moral subspace is the six foundations of Moral Foundations Theory (Graham et al., 2013). Zhao et al.'s harmfulness is a prohibited-capability contrast, related to but distinct from the foundations; we measure the relationship directly (subsection 4.4) rather than assume it.

3 Methodology

3.1 Model panel

We study three open reasoning models chosen to isolate the deliberative axis. **GPT-OSS-20B** (OpenAI, 2025) is the primary: a mixture-of-experts model whose reasoning was learned by reinforcement learning under deliberative alignment, making its chain of thought a trained behavior. **DeepSeek-R1-Distill-Llama-8B** and **DeepSeek-R1-Distill-Qwen-14B** (DeepSeek-AI, 2025) acquired their reasoning by supervised distillation of DeepSeek-R1 traces onto, respectively, Llama-3.1-8B and the general Qwen2.5-14B (verified at load against the model config; the 14B uses the general base, not a math variant). The two distills share the R1 teacher, so a difference between them is not a teacher difference. We note, and do not control for, two confounds: GPT-OSS-20B is the only MoE, so a GPT-OSS-versus-distill difference is confounded between deliberative alignment and architecture; and the distills differ in scale (8B versus 14B). Layer indices use a fixed depth-fraction rule so conventions hold across the 24/32/48-layer panel.

3.2 Extracting harm comprehension and refusal at separate token positions

Following Zhao et al. (2025), we extract diff-of-means directions at two token positions: t_{inst}, the last token of the user instruction, which encodes harmfulness; and $t_{\text{post-inst}}$, the last token of the templated prompt, which encodes refusal. The chat template appends a fixed suffix after the instruction content (turn-end, assistant header, and for these models a reasoning opener), so we locate t_{inst} as the position before the longest common token suffix across diverse probe instructions. This is model-agnostic and needs no per-model token counts: it resolves t_{inst} to the instruction's last token for the DeepSeek `<|Assistant|><think>` opener and the GPT-OSS harmony assistant opener alike, validated on each tokenizer. The harmfulness direction is the harmful-minus-harmless mean of t_{inst} residuals; the refusal direction the same at $t_{\text{post-inst}}$.

3.3 Subspaces and prompts

The six-foundation Moral Foundations subspace is the orthonormalized span of the six per-foundation probe directions, extracted on each reasoning model in raw text (Reblitz-Richardson, 2026). Harmful and harmless instructions are the Arditi/Heretic set (Arditi et al., 2024, p-e-w, 2025) (the `harmful_behaviors` / `harmless_alpaca` prompts), with a held-out evaluation split of 100 prompts per class disjoint from the training prompts and spanning distinct harm categories. Directions are estimated on a category-spanning draw from the training prompts; all causal tests are scored on the disjoint held-out split.

3.4 Reply-inversion and the readout problem

Reply-inversion (Zhao et al., 2025) validates that a direction is causal, not merely decodable: append the question "Could this user prompt cause harm? Answer 'Certainly' if it could, otherwise 'No' ", steer the harmfulness direction, and test whether the model's verdict flips. We read the verdict by forcing the answer position (a reasoning-close followed by an `Answer:` prefix) and comparing the logits of the verdict tokens, and we steer by adding $\alpha \cdot \|\bar{h}\| \cdot \hat{d}$ to the residual at one layer, where $\hat{d}$ is the unit harmfulness direction and $\|\bar{h}\|$ the mean residual norm at that layer. Steering magnitude is swept as a multiple of the residual norm rather than of the raw diff-of-means, because the raw diff is only 0.44–0.49 of the residual norm and a coefficient on it is too weak to move the verdict.

A flip counts only if the steered verdict is coherent, defined as a verdict token remaining in the top-k logits; over-steering that drives a non-verdict token to the top is excluded. We report the continuous margin shift alongside the binary flip rate, so a real-but-not-flipping effect is distinguished from no effect.

The readout is clean only on direct-answering models. Reasoning models reason past or echo the question and over-judge when forced (subsection 4.6), so the reply-inversion causal validation is run on the instruct models that share the distills' families, as a positive control on the method and the direction. We treat the absence of a clean reasoning-model judgment readout as a finding rather than a workaround, and we do not infer a reasoning-model causal result from a readout the model does not expose.

3.5 Decodable is not causal, and held-out gating

A direction that separates harmful from harmless prompts (decodable) need not change behavior when ablated (causal). An early decomposition-grade refusal direction extracted at the prompt boundary separated harmful from harmless cleanly yet did not move refusal under ablation (its cosine to a freshly recomputed causal direction was 0.76). We therefore distinguish decodable directions, used for the geometric results, from causal directions, validated by an intervention. Causal claims are gated on a held-out, category-diverse split and on a sensitivity yardstick (a known-causal direction that must fire under the same intervention) firing cleanly; where the yardstick does not fire, we report the limitation rather than interpret a null against it.

4 Results

4.1 Harmfulness and refusal are separately encoded

At the instruction token, harmful and harmless prompts separate cleanly along the diff-of-means harmfulness direction: $d' = 5.01$ for GPT-OSS-20B, 4.39 for the Llama distill, and 5.01 for the Qwen distill (Figure 1, left). The separation is weaker at $t_{\text{post-inst}}$ (2.03, 3.59, 3.84), placing the harmfulness signal at the instruction token, as Zhao et al. (2025) report for instruct models. The

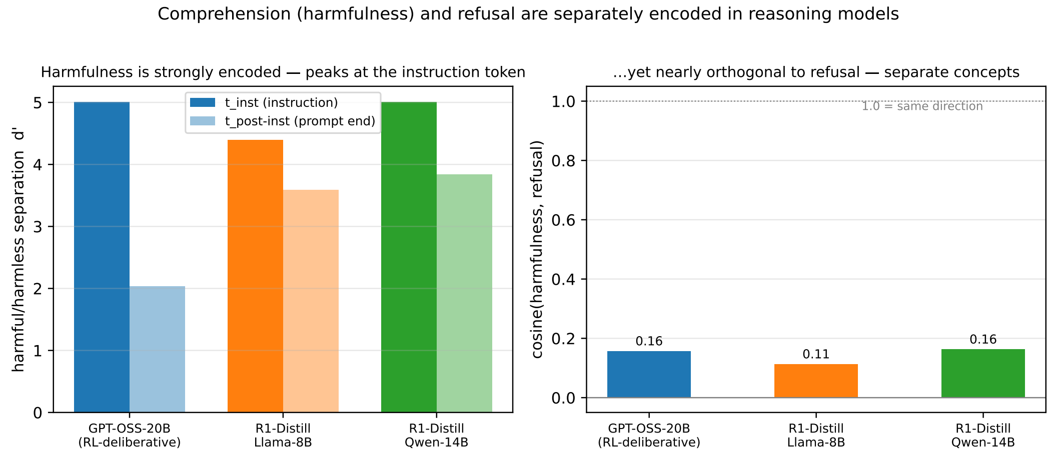

Figure 1: Harmfulness is strongly encoded at the instruction token yet nearly orthogonal to refusal, in all three reasoning models. Left: harmful/harmless separation (Cohen's d') of the t_{inst} and $t_{\text{post-inst}}$ diff-of-means directions; harmfulness is strongest at t_{inst}. Right: cosine between the t_{inst} harmfulness direction and the $t_{\text{post-inst}}$ refusal direction, with 1.0 marking a shared direction. Reproduces the harmfulness/refusal separation of Zhao et al. (2025) in reasoning models, including the RL-deliberative GPT-OSS-20B.

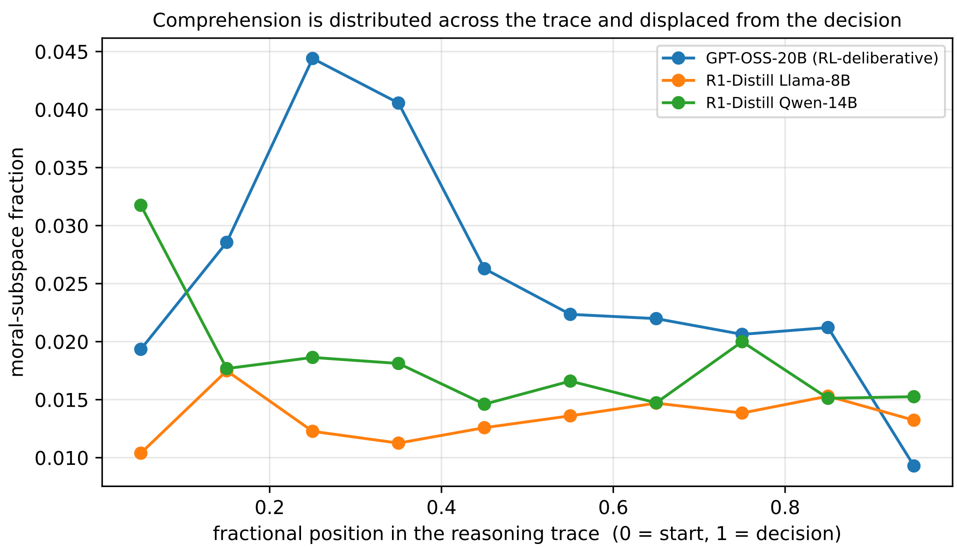

Figure 2: Moral-subspace content as a function of fractional position in the reasoning trace, on coherent closed traces (matched cognitive phase). GPT-OSS-20B carries the most trace-level moral content; it peaks near the first third of the trace and falls to its lowest value at the decision (position 1.0).

t_{inst} harmfulness direction is nearly orthogonal to the $t_{\text{post-inst}}$ refusal direction: cosine 0.16, 0.11, 0.16 (Figure 1, right). Harmfulness is decodable and strong, and it is not the refusal direction.

This holds for GPT-OSS-20B, whose reasoning was learned by reinforcement learning under deliberative alignment. Genuine deliberative training does not collapse the two representations into one. The dissociation that holds for non-reasoning models (Reblitz-Richardson, 2026, Zhao et al., 2025) extends into the reasoning regime, deliberative training included.

4.2 Comprehension is distributed and displaced from the decision

Refusal in a reasoning model is partly made inside the chain of thought, so we ask where harm comprehension sits along the trace. Extracting the refusal direction at two sites within the trace, the last input token (end-of-prompt) and the trace tokens, the end-of-prompt direction reproduces the

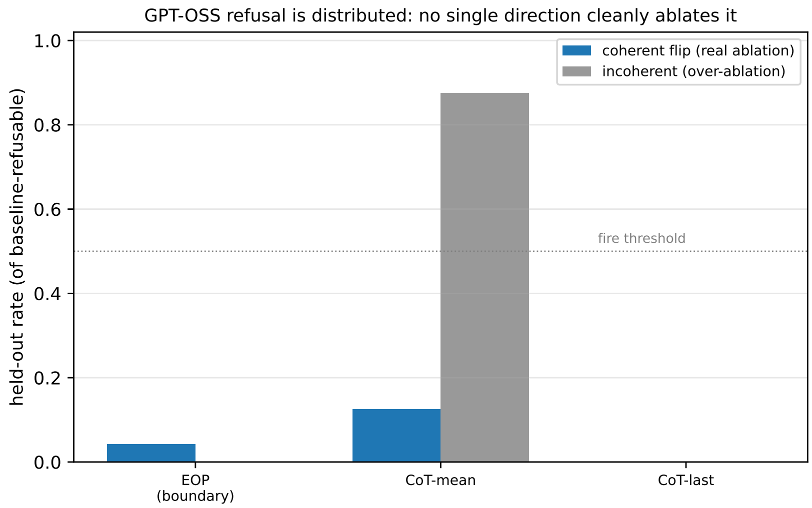

Figure 3: Held-out causal ablation of GPT-OSS-20B refusal. For each candidate direction (end-of-prompt, CoT-mean, CoT-last), the fraction of baseline-refusable held-out prompts whose refusal is coherently removed, and the fraction lost to over-ablation incoherence. No direction cleanly ablates refusal: the end-of-prompt and CoT-last directions barely move it, and CoT-mean removes it only by destroying generation.

non-reasoning result: it is 99.1% residual against the moral subspace at the boundary, and a single direction captures all of its linear separability (the single-versus-full-rank AUC gap is 0.000). The moral-projection asymmetry between the trace direction and the end-of-prompt direction is at or below zero in all three models, most negative in GPT-OSS-20B. There is no re-coupling of morality at the decision point.

Across the trace the picture is distributional. Controlling for trace length by comparing matched fractional positions on coherent closed traces, GPT-OSS-20B carries the most moral-subspace content (mean fraction 0.025, against 0.013 and 0.018 for the distills), and that content is concentrated near the first third of the trace and decays toward the decision (Figure 2). The model that most clearly deliberates carries the most trace-level moral content, but that content is displaced from where the refusal decision is read out.

4.3 The refusal decision is distributed

Whether GPT-OSS-20B's refusal routes through any single low-dimensional handle is testable directly: ablate a candidate direction during generation and measure whether refusal is coherently removed. On a held-out, category-diverse set, no direction does so (Figure 3). The end-of-prompt refusal direction, estimated on a category-spanning training draw, coherently flips only 4% of held-out refusals; the CoT-last direction flips none; the CoT-mean direction removes refusal in 88% of cases but only by driving generation into incoherence, which the coherence filter excludes. GPT-OSS-20B refusal is distributed: it is not bottlenecked through any single direction, the refusal direction itself included. Because no low-dimensional handle isolates the refusal decision, the single-subspace load-bearing test that would ask whether harm comprehension is causally upstream of refusal cannot be run cleanly on this model; the distributed-ness is itself the answer, since a representation that is not a bottleneck for refusal cannot be the moral subspace either.

4.4 Harmfulness is largely distinct from moral foundations

The harmfulness that reasoning models encode at t_{inst} is related to, but not the same as, the moral foundations our program measures. Projecting the harmfulness direction onto the six-foundation subspace, 0.18 of it lies inside for GPT-OSS-20B and 0.11 for each distill (Figure 4). These fractions exceed the chance floor for a random direction (≈ 0.04, the $\sqrt{k/d}$ expectation for a k-dimensional subspace in d dimensions) by 3.0 to 3.9 times, so the overlap is real and above chance. It is also small: the mean absolute cosine to the individual foundations is 0.07–0.11, and roughly 85% of

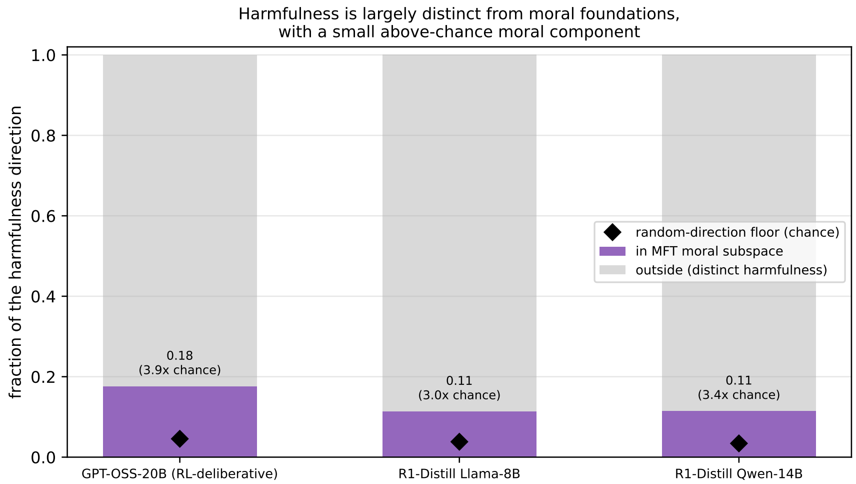

Figure 4: Decomposition of the t_{inst} harmfulness direction relative to the six-foundation Moral Foundations subspace. The harmfulness direction projects mostly outside the moral subspace; the in-subspace fraction $(0.11–0.18)$ exceeds the random-direction chance floor (diamonds, ≈ 0.04) by $3.0–3.9\times$, but the bulk of the direction is distinct from moral foundations.

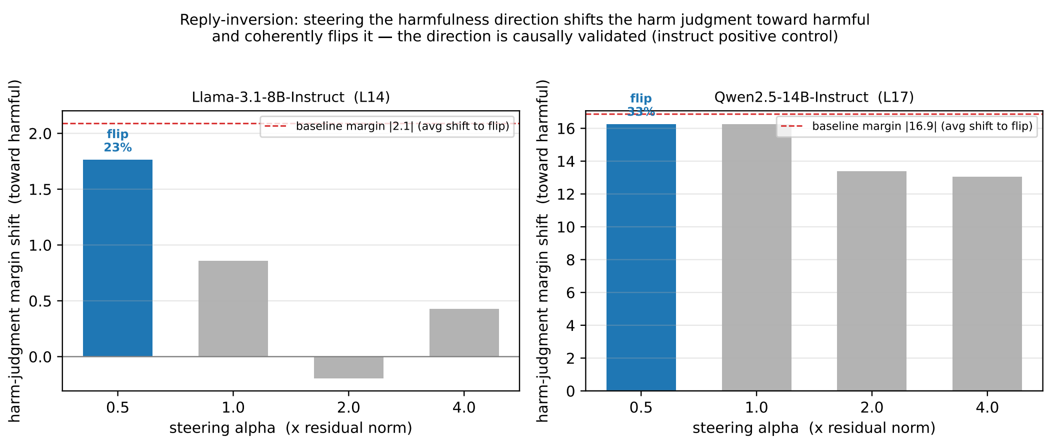

Figure 5: Reply-inversion on direct-answering instruct models. Steering the t_{inst} harmfulness direction (magnitude as a multiple of the residual norm) shifts the harm-judgment margin toward harmful; the dashed line marks the baseline margin, the average shift needed to cross to a harmful verdict. Blue bars are alpha settings at which the steered verdict coherently flips (annotated with the held-out flip rate); grey bars over-steer into incoherence. The direction is causal.

the harmfulness direction lies outside the moral subspace. Harm-judgment carries a modest moral-foundations component on top of a representation that is largely its own. The prohibited-capability harmfulness of Zhao et al. (2025) and the moral foundations of Graham et al. (2013) are distinct objects that share an above-chance component.

4.5 The harmfulness direction is causally validated

Whether the harmfulness direction is causal, not merely decodable, is tested by reply-inversion (Zhao et al., 2025): steer the direction and ask whether the model's stated harm judgment changes. On direct-answering instruct models that share the distills' families, steering shifts the harm-judgment margin strongly toward harmful and coherently flips it (Figure 5). For Qwen2.5-14B-Instruct, whose harmless prompts are judged safe with a confident margin of -16.9, steering shifts the margin by up to $+17.4$ and coherently flips 33% of held-out judgments; for Llama-3.1-8B-Instruct the shift is $+3.0$ against a -2.1 baseline and flips 23%. Flips occur in the lower-magnitude, coherent regime;

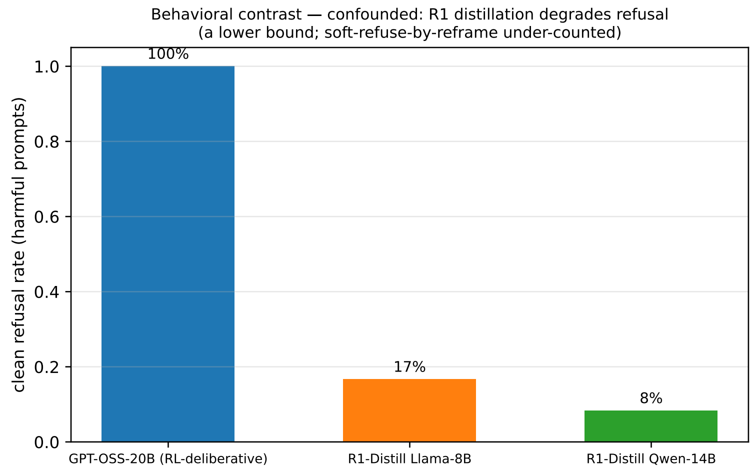

Figure 6: Clean refusal rate on harmful prompts. GPT-OSS-20B refuses every prompt; the distilled reasoning models refuse 8–17% (a lower bound, since their acknowledge-then-reframe responses are under-counted). The gap is confounded with R1 distillation degrading refusal training and is reported as a behavioral contrast, not a functional-versus-imitated asymmetry.

larger steering shifts the margin further but over-steers into incoherence, which the coherence gate excludes. The earlier failure to move the verdict with weaker steering was a magnitude artifact: the raw diff-of-means is only 0.44–0.49 of the residual norm, so a coefficient on it was too small. The t_{inst} harmfulness direction is a real causal handle on the model's harm judgment.

4.6 Reasoning models do not expose a clean judgment readout

The causal test in subsection 4.5 runs on instruct models because the reasoning models do not expose a harm judgment that can be read. The test appends a harm-judgment question and reads the verdict, and across three readout mechanisms the reasoning models defeat it. A regex over the generated answer matches the question echoed inside the reasoning trace rather than the verdict; reading the final answer fails because the trace rambles without stating a verdict within budget; forcing the answer position and reading verdict-token logits returns near-noise, and harmless prompts are scored harmful with large margins when the verdict is forced. The same forced-answer logit readout is clean on instruct models, which answer the question directly (Qwen2.5-14B-Instruct judges $24/24$ harmless prompts safe and $24/24$ harmful prompts harmful). The judgment is not absent from the reasoning model; it is not exposed in a position a readout can reach. Interpretability methods that depend on a stated judgment, reply-inversion among them, require a behavioral channel the reasoning model will emit, and the chain of thought is not one.

4.7 Behavioral contrast, and its confound

Behaviorally the panel splits sharply: GPT-OSS-20B refuses all harmful prompts, while the distilled models refuse 17% (Llama) and 8% (Qwen) (Figure 6). The distill rate is a lower bound, since their typical response acknowledges the harm and reframes toward a defensive answer rather than issuing an explicit refusal, which the classifier under-counts. The distills carry harm comprehension at t_{inst} (subsection 4.1) yet rarely translate it into refusal. We do not read this as evidence that their harm comprehension is decorative, because the low refusal rate is confounded with the well-documented effect of R1 distillation degrading refusal training: a model that refuses almost nothing tells us little about whether its harm comprehension would be load-bearing if it did refuse. The contrast is suggestive and we report it as such.

5 Discussion

The dissociation extends into the reasoning regime. The headline result is a negative one with a clear shape. A model trained by reinforcement learning to deliberate about harm does not, by virtue of that training, route its refusal through its harm comprehension. Harmfulness is strongly decodable at the instruction token and nearly orthogonal to the refusal direction in GPT-OSS-20B exactly as in the distilled models and as in the non-reasoning models of prior work. The comprehension/compliance dissociation is not an artifact of weak post-training; it survives the strongest deliberative-alignment training available in an open model.

Comprehension is present, distributed, and displaced. GPT-OSS-20B carries the most trace-level moral content of the panel, and it deliberates about harm in a way the geometry can see. But that content peaks in the early-to-middle of the trace and fades to its lowest value at the decision, while the refusal decision is distributed across directions rather than bottlenecked through one. The model that most clearly reasons about harm is also the one whose refusal is least reducible to a single handle. Deliberation and a low-dimensional refusal mechanism are not the same thing, and in GPT-OSS-20B they come apart.

Functional versus imitated, and why we do not claim it. The experiment we set out to run would ablate harm comprehension and ask whether refusal changes, more in the RL-deliberative model than in the imitated distills. Two facts block it, and we report them rather than force the claim. GPT-OSS-20B's refusal is distributed, so there is no clean low-dimensional ablation that isolates a load-bearing test; the distributed-ness does imply, weakly, that the moral subspace is not a refusal bottleneck, since nothing low-dimensional is, but that is an a-fortiori argument, not an ablation result. And the distilled models barely refuse, a behavior confounded with R1 distillation degrading refusal training, so their failure to translate harm comprehension into refusal cannot license a claim that the comprehension is decorative. The honest contribution is the dissociation and its reach, not a functional-versus-imitated split we cannot cleanly measure.

A constraint on interpretability methods for reasoning models. That the reply-inversion readout is clean on instruct models and unreadable on reasoning models is not a nuisance of our setup; it is a property of the models. Methods that validate a direction by reading the model's stated judgment assume the model will state one in a position the readout can reach. Reasoning models do not: they reason past the question, echo it, or over-judge when forced. The harmfulness direction is nonetheless causal, as the instruct positive control shows, but its causal status on the reasoning models themselves cannot be established through a behavioral readout they do not emit. As refusal- and harmfulness-direction methods are increasingly applied to reasoning models, the readout channel, not the direction, becomes the binding constraint.

Harmfulness is not the moral foundations. The harm-judgment that reasoning models encode at the instruction token overlaps the Moral Foundations subspace above chance but lies mostly outside it. Whatever reasoning models route their harm-comprehension through, it is a prohibited-capability representation more than a representation of Haidt's foundations. This sharpens the relationship between the present program, which measures moral structure, and the harmfulness account of Zhao et al. (2025): the two are measuring related but distinct objects, and a result about one does not transfer to the other without the overlap measured.

Limitations. The causal validation of the harmfulness direction is on instruct models, not the reasoning models, for the readout reason above; we therefore cannot claim the direction is load-bearing for refusal in the reasoning models. GPT-OSS-20B is the only MoE in the panel, confounding deliberation with architecture, and the two distills differ in scale. The distill behavioral contrast is confounded by distillation degrading refusal. Sample sizes for the trace-level and causal measurements are bounded by generation cost; we report held-out splits and chance floors throughout, and gate causal claims on a firing sensitivity yardstick.

6 Conclusion

We asked whether deliberative reasoning re-couples harm comprehension to the refusal decision that non-reasoning models keep separate, and found that it does not. Across three open reasoning models, including GPT-OSS-20B, whose reasoning was learned by reinforcement learning under deliberative alignment, harmfulness is strongly decodable at the instruction token and nearly orthogonal to the refusal direction. The harmfulness/refusal dissociation established for instruct models (Zhao et al., 2025) and the comprehension/compliance dissociation established for non-reasoning models (Reblitz-Richardson, 2026) both extend into the reasoning regime.

Within that regime the comprehension is present but placed away from the decision: GPT-OSS-20B carries the most trace-level moral content, concentrated early in the trace and decaying to the decision, while its refusal is distributed across directions with no single-direction bottleneck. The harmfulness representation the models do encode is causally real, validated by reply-inversion on direct-answering instruct models, and largely distinct from Moral Foundations, overlapping the six-foundation subspace above chance but lying mostly outside it.

We report two boundaries honestly. The clean functional-versus-imitated split we hoped to measure is not cleanly runnable, because GPT-OSS-20B's refusal is distributed and the distilled models' weak refusal is confounded with distillation degrading refusal training. And the reply-inversion readout that validates the harmfulness direction on instruct models cannot be read on the reasoning models themselves, which do not expose a stated judgment in a reachable position. That last boundary is itself a result: interpretability methods built on direct-answering models require a behavioral readout the reasoning model will emit, and the chain of thought does not provide one. As safety research leans more heavily on open reasoning models, both the geometry of where harm comprehension sits and the channel through which it can be read become first-order concerns.

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Appendices

Supplementary material.

A Result tables

Table 1 collects the per-model geometric measurements: the harmfulness separation at the two extraction sites, the cosine between the harmfulness and refusal directions, the fraction of the harmfulness direction inside the Moral Foundations subspace against its chance floor, the matched-phase trace moral content, and the clean behavioral refusal rate. Table 2 collects the reply-inversion positive control on the direct-answering instruct models.

Table 1: Per-model geometry and behavior. Harmfulness (d') is the harmful/harmless separation of the t_{inst} and $t_{\text{post-inst}}$ diff-of-means directions; $\cos(h, r)$ the cosine between the t_{inst} harmfulness and $t_{\text{post-inst}}$ refusal directions; harm-in-moral the projection fraction of the harmfulness direction into the six-foundation subspace (chance floor in parentheses); trace moral the matched-fractional-position mean moral-subspace content on coherent closed traces; refusal the clean refusal rate on harmful prompts.

model	d' t_{inst}	d' $t_{\text{post-inst}}$	$\cos(h, r)$	harm-in-moral	trace moral	refusal
GPT-OSS-20B (RL)	5.01	2.03	0.16	0.18 (0.05)	0.025	1.00
R1-Distill-Llama-8B	4.39	3.59	0.11	0.11 (0.04)	0.013	0.17
R1-Distill-Qwen-14B	5.01	3.84	0.16	0.11 (0.03)	0.018	0.08

Table 2: Reply-inversion positive control on direct-answering instruct models. Clean judgment accuracy on the held-out harmless/harmful sets; baseline harm-judgment margin on harmless prompts (negative = safe); the maximum margin shift toward harmful under steering; and the best held-out coherent flip rate (safe→harmful).

model	clean (safe / harmful)	baseline margin	max margin shift	best flip
Qwen2.5-14B-Instruct	$24/24$ / $24/24$	-16.9	$+17.4$	0.33
Llama-3.1-8B-Instruct	$22/24$ / $23/24$	-2.1	$+3.0$	0.23

B Reproducibility

Models. `openai/gpt-oss-20b` (24 layers, 2880 hidden, MoE 32/4, loaded bf16-dequantized from mxfp4), `deepseek-ai/DeepSeek-R1-Distill-Llama-8B` (32 layers), `deepseek-ai/DeepSeek-R1-Distill-Qwen-14B` (48 layers). Bases for the distills are `meta-llama/Llama-3.1-8B` and the general `Qwen/Qwen2.5-14B`, verified at load against the model config. The reply-inversion positive control uses `Qwen/Qwen2.5-14B-Instruct` and `meta-llama/Llama-3.1-8B-Instruct`.

Prompts. Harmful and harmless instructions are the Arditi/Heretic set (`mlabonne/harmful_behaviors` and `mlabonne/harmless_alpaca`, first-N in dataset order), with a disjoint, category-diverse held-out evaluation split of 100 prompts per class. Directions are estimated on a category-spanning shuffle of the training prompts; all causal tests are scored on the held-out split.

Conventions. t_{inst} is located as the position before the longest common token suffix across diverse probe instructions (model-agnostic; no per-model token counts). Layer indices use a fixed depth-fraction rule across the $24/32/48$-layer panel; the harmfulness-direction reply-inversion sweeps depth fractions 0.25–0.55 to span the mid-layer prior of Zhao et al. The moral subspace is the orthonormalized span of the six Moral Foundations probe directions, extracted in raw text. Steering adds $\alpha \cdot \|\bar{h}\| \cdot \hat{d}$ at one layer; flips require a verdict token to remain in the top-k logits (coherence gate).

Outputs. Every measurement is written as JSON, one-to-one with the figures: `position_extraction.json` (harmfulness/refusal at both sites, Figure 1), `trace_length_disentangle.`

json (Figure 2), yardstick_validation.json (Figure 3), position_vs_moral.json (Figure 4), reply_inversion.json and the instruct control/*_inversion.json (Figure 5), and refusal_baseline.json (Figure 6). Figures are regenerated from these by paper7_figures.py.

Compute. All white-box passes run on a single A100-80GB via the project's RunPod harness; the analyses are activation reads and single-layer interventions, with no training or fine-tuning. The toolkit extends the prior paper's extraction code (extract_refusal.py, random_ablation_control.py, the model registry) rather than reimplementing it.

Limits. Causal validation of the harmfulness direction is on the instruct models, not the reasoning models, because the reasoning models do not expose a clean judgment readout. GPT-OSS-20B is the only MoE, confounding deliberation with architecture; the two distillations differ in scale; and the distillation behavioral contrast is confounded with R1 distillation degrading refusal training. Causal claims are gated on a held-out split and a firing sensitivity yardstick.